

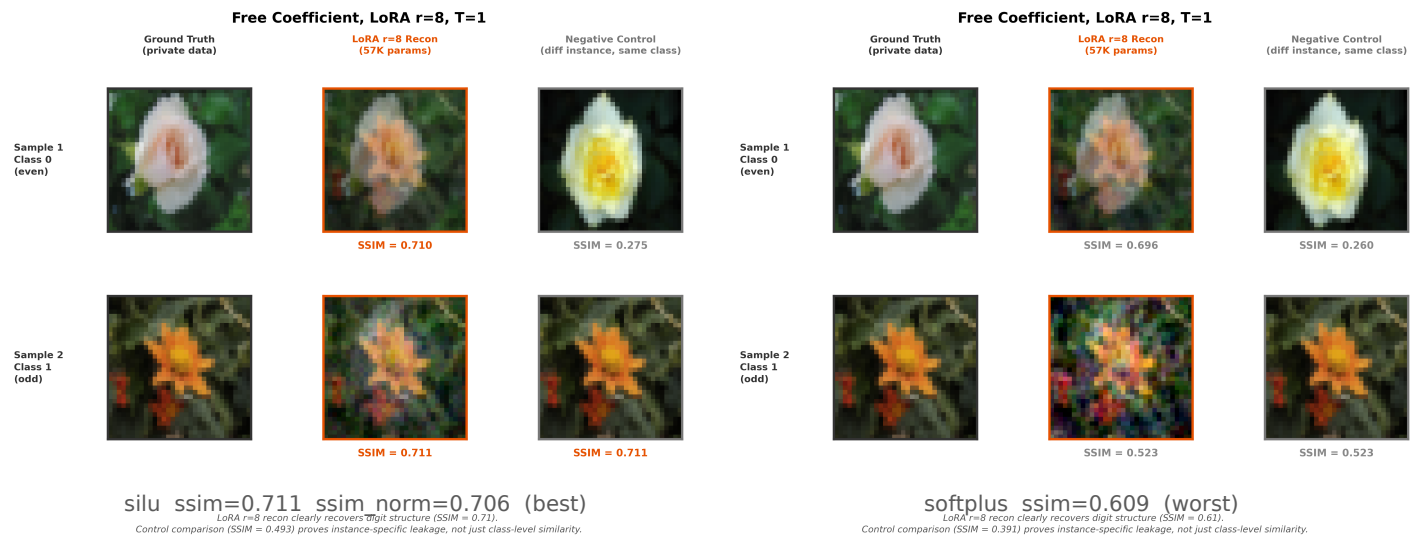
Flowers-Native LoRA Reconstruction

Realistic free-coefficient attack — worked examples

Native RGB flowers (32x32, D=3072), LoRA r=8, T=1, N=2 unless noted. REALISTIC free-coefficient attack (coefficients optimized with x, like Haim et al.), verified recipe: SGD extraction + relu_alpha=10000 (~ReLU) + consistency_weight=1.0 + restarts. Each grid: Ground Truth (top) / LoRA reconstruction (middle) / Negative control = different instance of the same species (bottom). All examples are the actual saved tensors in results/.

1. Activation ranking (the headline flip)

Under the realistic fixed-ReLU attacker the ranking is silu > gelu > mish > elu > softplus — the OPPOSITE of the oracle 'softplus wins'. silu/gelu (what real ViTs use) leak MOST. This is robust across ssim / ssim_norm / ssim11 / ncc / l2 (silu top, softplus bottom under every metric).

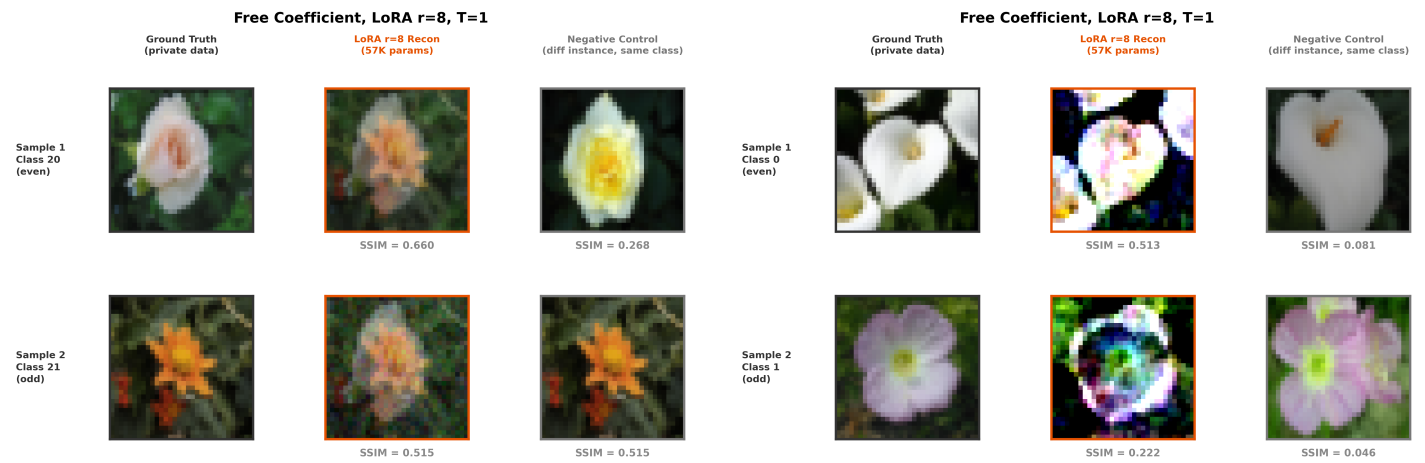


Metric table (free-c, r8, N=2):

silu	ssim 0.711	ssim_norm 0.706	ssim11 0.741	ncc 1442(low=best)	l2 5.60
gelu	ssim 0.692	ssim_norm 0.687	ssim11 0.719	ncc 1657	l2 5.80
mish	ssim 0.689	ssim_norm 0.684	ssim11 0.721	ncc 1730	l2 5.67
elu	ssim 0.643	ssim_norm 0.645	ssim11 0.694	ncc 1929	l2 5.77
softplus	ssim 0.609	ssim_norm 0.611	ssim11 0.667	ncc 2299	l2 5.92

2. Q-B: seen (overlap) vs novel (held-out species)

theta_0 was trained WITHOUT species 0-19. 'seen' = fine-tune on species it knows; 'novel' = the held-out ones. Raw SSIM: seen 0.587 > novel 0.368 ('familiarity aids leakage'). BUT this is a metric artifact: novel is 50% CLIPPED (its reconstruction blows past [0,1]) — visible below as the washed-out / saturated look. Under scale-invariant NCC the novel STRUCTURE is actually closer.



seen ssim=0.587 clip=0.00	novel ssim=0.368 clip=0.50 (artifact!)
LoRA r=8 recon clearly recovers digit structure (SSIM = 0.59).	LoRA r=8 recon clearly recovers digit structure (SSIM = 0.37).
Control comparison (SSIM = 0.391) proves instance-specific leakage, not just class-level similarity.	Control comparison (SSIM = 0.063) proves instance-specific leakage, not just class-level similarity.

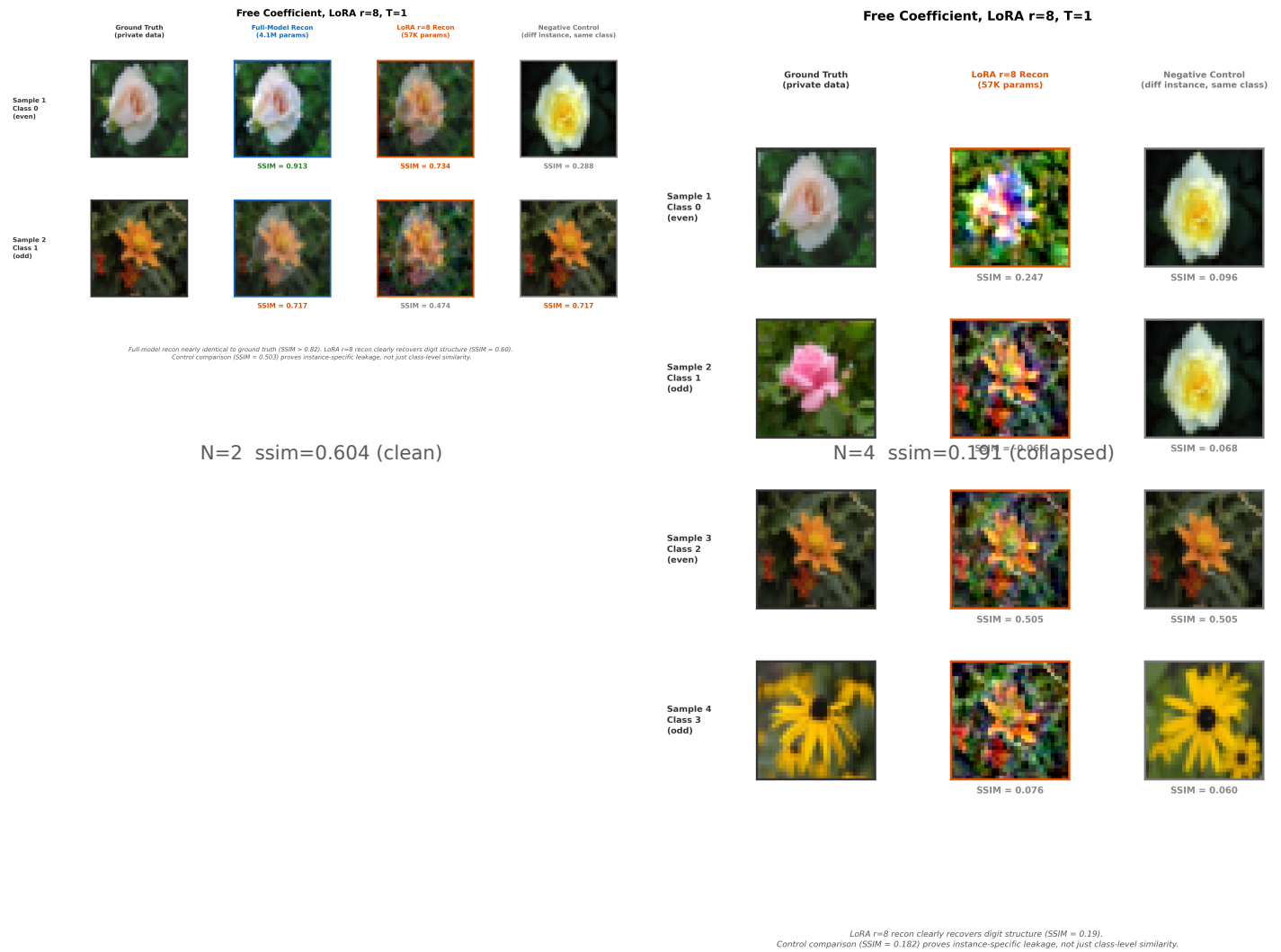
seen ssim 0.587 ssim_norm 0.580 ncc 5118 l2 6.99 clip 0.00

novel ssim 0.368 ssim_norm 0.495 ncc 413 l2 19.34 clip 0.50 <- clipping tanks raw SSIM

Caveat: the seen>novel MAGNITUDE is inflated by novel's clipping (the soft [-1,1] box gap). Report Q-B on ssim_norm/NCC (scale-robust) or fix the pixel box before quoting the gap.

3. The N>2 superposition wall

N=2 reconstructs cleanly (ssim 0.604). At N=4 the joint optimization collapses to superpositions (ssim 0.191 ~ control) — the images blend. Structural separators (SVD/ICA/SPEAR) need the FULL gradient and fail on the raw LoRA adapter (its SVD spans the adapter subspace, not the data) — which is a concrete motivation for the Gradient Bridge. x-space methods (peeling) help modestly.



4. Why SSIM alone is not enough (metric robustness)

- Activation ranking: ROBUST — silu top / softplus bottom under ssim, ssim_norm, ssim11, ncc, l2. Safe to headline.
- Q-B seen>novel: FRAGILE — raw SSIM inflated by novel's 50% clipping; scale-invariant NCC partly reverses it. Report scale-robust metrics.
- Rank curve $r=4..64$: consistent U-shape across metrics.

Metric roles: raw SSIM = perceptual but background/clipping-dominated; ssim_norm/NCC = structure only (scale-invariant); L2 = scale-sensitive; control-margin (sample - control) = instance leakage; retrieval (top-1 vs 1/N) = instance identification. Leakage claims rest on control-margin + retrieval, not raw SSIM.